

EXPERIMENT – 2 GO FURTHER – 2

Objective

Digital Password Lock System using Arduino UNO, OLED Display, Keypad, LEDs, and Buzzer

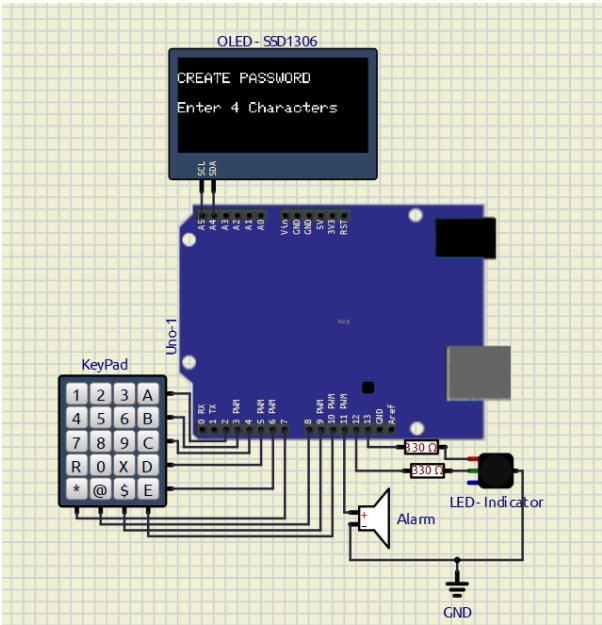
Problem Statement

To design and implement a Digital Password Lock System using Arduino UNO that allows a user to create a 4-character password, authenticate access using the password, provide visual and audio feedback for successful and failed login attempts, and support secure password reset functionality.

Components Required

Sl. No.	Component	Quantity
1	Arduino UNO	1
2	SSD1306 OLED Display (I2C)	1
3	4×5 Matrix Keypad	1
4	Red LED	1
5	Green/Blue LED	1
6	Buzzer	1
7	Breadboard	1
8	Connecting Wires	As Required
9	USB Cable	1

Circuit Diagram



Connections

OLED Display

OLED Pin	Arduino UNO
VCC	5V
GND	GND
SDA	A4
SCL	A5

Keypad Connections

Keypad Pin	Arduino Pin
R1	D2
R2	D3
R3	D4
R4	D5
R5	D6
C1	D7
C2	D8
C3	D9
C4	D10

LED and Buzzer Connections

Component	Arduino Pin
Buzzer	D11
Green LED	D12
Red LED	D13

Code

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <Keypad.h>

// ===== OLED =====
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

// ===== PINS =====
#define BUZZER 11
#define BLUE_LED 12
#define RED_LED 13

// ===== KEYPAD =====
const byte ROWS = 5;
const byte COLS = 4;

char keys[ROWS][COLS] = {
  {'1', '2', '3', 'A'},
  {'4', '5', '6', 'B'},
  {'7', '8', '9', 'C'},
  {'R', '0', 'X', 'D'},
  {'*', '@', '$', 'E'}
};

byte rowPins[ROWS] = {2, 3, 4, 5, 6};
byte colPins[COLS] = {7, 8, 9, 10};

Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);

// ===== VARIABLES =====
String adminPassword = "";
int attemptsLeft = 3;

// =====

void showMessage(String line1, String line2 = "") {

  display.clearDisplay();
```

```
display.setTextSize(1);
display.setTextColor(WHITE);

display.setCursor(0, 10);
display.println(line1);

display.setCursor(0, 30);
display.println(line2);

display.display();
}

// =====

bool isValidPassword(String pass) {

    // Only 4 character password allowed

    if (pass.length() != 4)
        return false;

    return true;
}

// =====

String readPassword() {

    String pass = "";

    while (pass.length() < 4) {

        char key = keypad.getKey();

        if (key) {

            tone(BUZZER, 1500, 80);

            pass += key;

            display.clearDisplay();

            display.setTextSize(1);
            display.setTextColor(WHITE);

            display.setCursor(0, 10);
            display.print("ENTER PASSWORD");

            display.setCursor(0, 30);

            for (int i = 0; i < pass.length(); i++) {
                display.print("*");
            }
        }
    }
}
```

```
        display.display();
    }
}

delay(500);

return pass;
}

// =====

void accessGranted() {

    digitalWrite(BLUE_LED, HIGH);

    tone(BUZZER, 2000, 300);

    showMessage("ACCESS GRANTED");

    delay(3000);

    digitalWrite(BLUE_LED, LOW);
}

// =====

void accessDenied() {

    digitalWrite(RED_LED, HIGH);

    tone(BUZZER, 500, 500);

    showMessage(
        "WRONG PASSWORD",
        "Attempts Left: " + String(attemptsLeft)
    );

    delay(2000);

    digitalWrite(RED_LED, LOW);
}

// =====

void alarmMode() {

    showMessage("SYSTEM LOCKED", "ALARM ACTIVE");

    while (1) {

        digitalWrite(RED_LED, HIGH);
        tone(BUZZER, 700);
```

```
        delay(300);

        digitalWrite(RED_LED, LOW);
        noTone(BUZZER);

        delay(300);
    }
}

// =====

void createPassword() {

    while (1) {

        showMessage(
            "CREATE PASSWORD",
            "Enter 4 Characters"
        );

        String newPass = readPassword();

        if (isValidPassword(newPass)) {

            adminPassword = newPass;

            tone(BUZZER, 2000, 300);

            showMessage("PASSWORD SAVED");

            delay(2000);

            break;
        }

        else {

            tone(BUZZER, 400, 1000);

            showMessage("INVALID PASSWORD");

            delay(2000);
        }
    }
}

// =====

void resetPassword() {

    showMessage("RESET MODE", "Enter Old Pass");

    String oldPass = readPassword();
```

```
// Correct old password
if (oldPass == adminPassword) {

    tone(BUZZER, 2000, 300);

    showMessage("CORRECT", "Set New Password");

    delay(2000);

    while (1) {

        String newPass = readPassword();

        if (isValidPassword(newPass)) {

            adminPassword = newPass;

            tone(BUZZER, 2000, 500);

            showMessage("PASSWORD RESET");

            delay(2000);

            return;
        }

        else {

            tone(BUZZER, 400, 1000);

            showMessage("INVALID PASSWORD");

            delay(2000);
        }
    }
}

// Wrong old password
else {

    tone(BUZZER, 300, 1000);

    showMessage("WRONG OLD PASS");

    delay(2000);
}
}

// =====

void setup() {

    pinMode(BUZZER, OUTPUT);
```

```
pinMode(BLUE_LED, OUTPUT);
pinMode(RED_LED, OUTPUT);

Wire.begin();

display.begin(SSD1306_SWITCHCAPVCC, 0x3C);

display.clearDisplay();
display.display();

// Create admin password
createPassword();
}

// =====

void loop() {

  showMessage(
    "ENTER PASSWORD",
    "Attempts Left: " + String(attemptsLeft)
  );

  String enteredPassword = "";

  while (enteredPassword.length() < 4) {

    char key = keypad.getKey();

    if (key) {

      // Reset password mode
      if (key == 'R') {

        resetPassword();

        return;
      }

      tone(BUZZER, 1500, 80);

      enteredPassword += key;

      display.clearDisplay();

      display.setTextSize(1);
      display.setTextColor(WHITE);

      display.setCursor(0, 10);
      display.print("ENTER PASSWORD");

      display.setCursor(0, 30);

      for (int i = 0; i < enteredPassword.length(); i++) {
```

```
        display.print("*");
    }

    display.display();
}

// ===== CHECK PASSWORD =====

if (enteredPassword == adminPassword) {

    attemptsLeft = 3;

    accessGranted();
}

else {

    attemptsLeft--;

    accessDenied();

    if (attemptsLeft <= 0) {

        alarmMode();
    }
}
}
```

Output

System Startup

OLED Display:

```
SET ADMIN PASS
Enter 4 Characters
```

Password Saved

OLED Display:

```
PASSWORD SAVED
System Ready
```

User Login

OLED Display:

ENTER PASSWORD
Unlock The System

Correct Password

OLED Display:

ACCESS GRANTED

- Green LED turns ON.
- Buzzer beeps.

Wrong Password

OLED Display:

ACCESS DENIED
LEFT: 2

- Red LED turns ON.
- Remaining attempts decrease.

Alarm Mode

OLED Display:

SYSTEM LOCKED
PRESS R RESET

- Red LED blinks continuously.
- Buzzer sounds continuously.

Password Reset

OLED Display:

RESET PASSWORD
Enter Old Pass

After successful verification:

```
CORRECT PASSWORD
Set New Pass
```

Problems Faced

1. OLED display address mismatch (0x3C / 0x3D).
 2. Incorrect SDA and SCL wiring.
 3. Keypad row and column mapping errors.
 4. Undefined LED pin names during coding.
 5. OLED text overflow due to long messages.
 6. Alarm mode creating an infinite loop.
 7. Debugging password reset functionality.
 8. Ensuring proper synchronization between keypad input and OLED display updates.
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Conclusion

The Digital Password Lock System was successfully implemented using Arduino UNO. The system allows secure password-based access control using a keypad and OLED display. Visual and audio feedback improve user interaction, while the three-attempt security mechanism and alarm mode enhance system security. The password reset feature further increases usability and flexibility. This experiment demonstrates practical applications of embedded systems in electronic security and access control.

Resources

11. Web Resources / References

1. SSD1306 OLED Display Datasheet <https://cdn-shop.adafruit.com/datasheets/SSD1306.pdf>
2. Matrix Keypad Working Principle <https://www.electronicsforu.com/technology-trends/learn-electronics/keypad-interfacing-arduino>
3. I2C Communication Protocol Overview <https://lastminuteengineers.com/i2c-lcd-arduino-tutorial/>
4. Password-Based Access Control Systems <https://circuitdigest.com/microcontroller-projects/password-based-door-lock-system-using-arduino>
5. Matrix Keypad Scanning Technique <https://www.geeksforgeeks.org/keypad-interfacing-with-arduino/>
6. Buzzer and Alarm Circuit Concepts <https://components101.com/modules/buzzer-module>